

Matrix inequalities and norms

Spectral norms of independent entries with regular moment growth

Difficulty: challenging **Importance:** interesting to the community**Status:** Lean verified**Topic:** Expected spectral norms of inhomogeneous random matrices

Lean verified — 2026-09-14. Formalization: Diar Heidary, unaffiliated. The upper comparison displayed below, which is the whole of the conjecture stated in this entry, is proved in Lean 4 and machine-checked. The proof source, its mechanical verification and its transitive axiom report are recorded in [Lean proof and verification evidence](#) below. The conjecture remains credited to Łatała and Świątkowski, and the reverse inequality quoted in this entry remains their Theorem 4.1 and is not formalized. AI assistance is disclosed; no external human peer review is claimed.

Rating rationale: The missing upper estimate must account for localized random fluctuations across all variance profiles. Even weighted random signs retain an iterated-logarithm gap. A sharp estimate would improve quantitative understanding of the largest singular value in random matrix models.

Statement

For a real random variable Z and $r > 0$, write $\|Z\|_{L_r} = (\mathbb{E}|Z|^r)^{1/r}$. Fix $\alpha \geq 1$. Let $X = (X_{ij}) \in \mathbb{R}^{n \times n}$ have independent mean-zero entries with finite absolute moments of every order, satisfying

$$\|X_{ij}\|_{L_{2r}} \leq \alpha \|X_{ij}\|_{L_r} \quad (1 \leq i, j \leq n, r \geq 1).$$

Set $[n] = \{1, \dots, n\}$ and

$$M(X) = \max_i \left(\sum_j \mathbb{E} X_{ij}^2 \right)^{1/2} + \max_j \left(\sum_i \mathbb{E} X_{ij}^2 \right)^{1/2},$$

$$D(X) = \max_{1 \leq k \leq n} \min_{\substack{I \subseteq [n] \\ |I| \leq k}} \sup_{\substack{s, t \in \mathbb{R}^n \\ \|s\|_2, \|t\|_2 \leq 1}} \left\| \sum_{\substack{i \in I \\ j \notin I}} X_{ij} s_i t_j \right\|_{L_{\ln(k+1)}}.$$

Conjecture (Łatała–Świątkowski). For every $\alpha \geq 1$ there is a constant $C_\alpha > 0$ such that, for every $n \geq 1$ and every such matrix,

$$\mathbb{E}\|X\|_2 \leq C_\alpha (M(X) + D(X)),$$

where $\|X\|_2$ is the spectral norm. The constant is independent of the dimension and entry laws. The same deterministic index set I deletes both rows and columns; the supremum is outside the random-variable moment. The definition includes $L_{\ln 2}$ when $k = 1$, using the displayed moment functional even though its exponent is below one.

The reverse inequality up to a constant depending only on α is proved, so the target is equivalent to the source's two-sided comparison. No identical-distribution or symmetry assumption is imposed on the entries.

Lean proof and verification evidence — 2026-09-14

Proof source, toolchain and pinned dependencies

The formalization is the Lean 4 project [DiarHaidary/Spectral-norms-of-independent-entries-with-regular-moment-growth](#) at the immutable revision [762bd5ec5050a96f5e6ba3926b6cda4816fcd4b0](#). That exact commit is registered with the Palomar registry as [PALOMAR-2026-09-14-000006, version 1](#), which also preserves an archival fork of it.

- Lean toolchain: *leanprover/lean4:v4.33.0*.
- *mathlib* pinned at [db584cd6d46c92f209a44c0f1c829460d327499d](#); the remaining eight Lake dependencies are pinned to exact commits in the committed *lake-manifest.json*.
- *Challenge.lean* imports *Mathlib* only.

Compared declaration and its correspondence to this entry

Comparator selects the single declaration *MI32.main_upper*, with *definition_names* empty, so no part of the statement is a replaceable hole. Its literal Lean text is in [Challenge.lean](#), the file a reader is expected to audit. In the notation of this entry it asserts

$$\forall \alpha \geq 1 \quad \exists C > 0 \quad \text{such that} \quad \mathbb{E}\|X\|_2 \leq C(M(X) + D(X))$$

for every $n \geq 1$ and every admissible X . The correspondence between the displayed quantities and the definitions in that file is:

- $\|Z\|_{L_r}$ is *moment*, defined as $(\int |Z|^r)^{1/r}$ with a *real* exponent. It is therefore the displayed functional also when $r < 1$, so the order $\ln 2$ at $k = 1$ is kept literally and is not replaced by one.
- The hypotheses on X are *RegularEntries*: entry measurability; joint independence of the family indexed by ordered pairs, as *iIndepFun*; each entry has Bochner integral zero; $|X_{ij}|^p$ is integrable for every real $p > 0$; and $\|X_{ij}\|_{L_{2r}} \leq \alpha \|X_{ij}\|_{L_r}$ for every real $r \geq 1$. Neither identical distribution nor symmetry is assumed.
- $\|X\|_2$ is *spectralNorm*, the operator norm of the Euclidean linear map of the matrix.
- $M(X)$ is *varianceScale*, the supremum of the row standard deviations plus the supremum of the column standard deviations.
- The inner supremum over $\|s\|_2, \|t\|_2 \leq 1$ after deleting I is *weakMoment*: a real supremum, over deterministic test vectors, of the scalar moment of *deletedBilinear*. That bilinear form sums over *Finset.univ* $\setminus I$ on both axes, so one and the same deterministic set deletes rows and columns, and the supremum sits outside the random-variable moment.
- $D(X)$ is *deletionScale*, the supremum over $1 \leq k \leq n$ of the infimum, over index sets of cardinality at most k , of *weakMoment* at the order $\ln(k + 1)$.
- The conjectured inequality is *UpperBoundAt*, quantified over every $n \geq 1$, every type, every measurable space on it, every probability measure and every such X . The constant is bound before all of those, so it depends only on α and not on the dimension or the entry laws.

The probability space is an arbitrary type rather than a finite one, and no estimate enters as a hypothesis or as a custom axiom. The constant is explicit: *SymmetricDeletion.deletionConstant* evaluated at 2α , times e . It is conservative and is not claimed to be sharp.

The hypothesis class is not vacuous. *MI32.RegularWitness.exists_regularEntries* exhibits an independent fair-sign matrix satisfying the hypotheses at $\alpha = 1$ with $M(X) > 0$, so the displayed bound is a genuine constraint rather than a statement about an empty class.

Verification record

This catalog **reviewed a public verification record**; it did not rerun the checks itself. [Independent maintainer review and authenticated evidence](#) record the statement checks and the public run inspected

for this catalog. Palomar’s mechanical verification of the registered commit ran on Palomar’s own Linux infrastructure on 2026-09-14T14:07:31Z:

- [workflow run 34852526384](#) in *PalomarRegistry/PalomarSubmission*;
- Comparator at [575674928e239f5bc452aab72d1dd7b0f1326494](#), checking that the Solution declaration proves the Challenge statement in a separate environment;
- an independent NanoDa kernel replay at [68d5ca9db226849b41a6fff59d796ff19d0a8840](#);
- *lean4export* at [15f6055e299ad5b89345e533cc2192f4cc00f659](#);
- the *landrun* sandbox at [811cff51ceaf3d9843708aa6d22e9b84ccac8b4](#).

The registry entry records trust level *high*, and its AI editorial review returned outcome *neutral*, meaning that no blocking problem was identified.

The author’s own local checks are reproducible from the repository root with:

```
lake exe cache get
lake build
lake env lean Audit.lean
lake build Challenge Solution
lake env lean FullTargetAudit.lean
python3 scripts/verify.py
```

The last of these exits zero only when the whole local Lean gate passes, and writes a dated record to *verification/status.json*. A successful build is not on its own evidence of the result; the Comparator and kernel-replay run above is.

Transitive axiom report

```
'MI32.main_upper' depends on axioms:
[propxt, Classical.choice, Quot.sound]
```

produced by *lake env lean FullTargetAudit.lean* and retained at *verification/full-target-axioms.log*. The Comparator configuration permits only those three axioms and the mechanical run enforced that. There is no *sorryAx*, no custom axiom and no reliance on native execution. *Challenge.lean* retains a deliberate *sorry*, which is the Challenge hole by construction and is never imported by the proof.

Scope, attribution and limits

- **Scope.** Only the upper comparison displayed in this entry is formalized. The reverse inequality, Theorem 4.1 of Latała–Świątkowski, is quoted in this entry for context and is not formalized or selected. No quantile or two-sided statement is claimed, and no claim is made about the sharpness of the constant’s dependence on α .
- **Mathematical attribution.** The conjecture is Conjecture 4.3 of Latała–Świątkowski. The proof argument and its formalization are by Diar Heidary. The manuscript is [MI32.pdf](#), with its [LaTeX source](#); its formal-verification section tabulates, statement by statement, which Lean declaration carries each numbered result.
- **Deliberate divergences from the manuscript**, recorded in the project’s *docs/SOURCE_FIDELITY.md*: the formal far-remainder estimate avoids the variance-envelope input of Latała–van Handel–Youssef by raising the screening threshold and using an elementary centered-Gram second moment, and the formal decomposition runs on the original rectangular matrix rather than on the matrix-symmetric dilation.
- **AI assistance is disclosed.** The proof, the Lean development and the exposition were produced with substantial AI-agent assistance under the author’s direction. The statement-fidelity review and the registry’s editorial review are AI reviews. Neither is human peer review and none is claimed. Registration with Palomar is not acceptance, endorsement or a novelty claim.

Known cases and numerical significance

The source proves the Gaussian-mixture case under the same moment condition. Weighted signs $X_{ij} = a_{ij}\varepsilon_{ij}$, with independent fair $\varepsilon_{ij} \in \{-1, 1\}$, satisfy the condition with $\alpha = 1$ and form the source's earlier Conjecture 1.2. Latała proves this case when $a_{ij} \in \{0, 1\}$ and proves the general weighted-sign estimate with an additional factor of order $\log \log \log n$. Meller's 2026 result gives an iterated-logarithm loss for a wider class of symmetric entries; it does not remove that loss. These cases remain grouped in this entry.

The spectral norm is the largest singular value and measures the largest Euclidean amplification by a random matrix. The formula combines row and column variance scales with moments of bilinear forms after deleting a limited number of coordinates, accounting for concentration on small parts of a matrix that simpler variance-only bounds can miss.

References and status check

- R. Latała and W. Świątkowski, *Norms of randomized circulant matrices*, Electronic Journal of Probability 27 (2022), paper 80, 1–23. DOI; current arXiv v2, dated 2022-05-27. Conjecture 4.3 on preprint p.25, with condition (25) on p.24, states the target. Theorem 4.1 supplies the lower bound; Proposition 4.4 and the paragraph before it cover Gaussian mixtures. Conjecture 1.2 on p.2 is the weighted-sign special case.
- R. Latała, *On the spectral norm of Rademacher matrices*. DOI; current arXiv v2, dated 2025-08-18. Equations (1.2)–(1.3), Theorem 1.1, and Theorem 1.9 give the weighted-sign conjecture and the stated partial results. This paper uses the comparable truncated logarithm $\max\{1, \ln k\}$ in place of $\ln(k+1)$.
- R. Meller, *Spectral norm of matrices with independent entries up to polyloglog*, arXiv:2512.23673v2, dated 2026-01-29, introduction, equation (3), and Theorem 1.1. It explicitly identifies the remaining logarithmic gap and Conjecture 4.3.

On 2026-09-11, checked the original paper's current version, both later papers, and exact-title, author/conjecture, Rademacher spectral-norm, proof/counterexample and 2026 searches. No full resolution was found. Gaussian operator-norm results and the general Bernoulli-process theorem do not prove this displayed formula. This is a bounded literature check.